Grad Refactor AI — Fine-Tuning & Evaluation Accuracy Report

Multi-Agent Python Refactoring System (`Qwen/Qwen2.5-Coder-3B-Instruct` + Native fp16 LoRA Adapter)

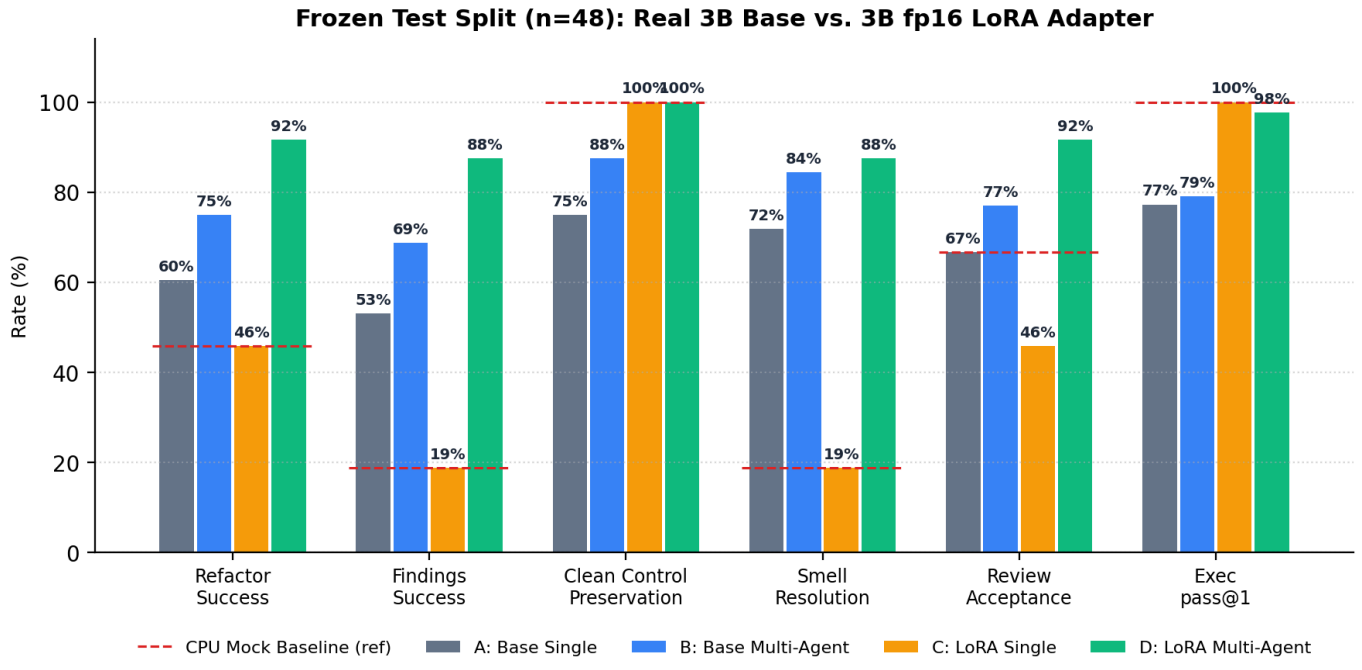
Generated: 2026-11-10T21:59:01Z • Test Split: `data/curated.jsonl` (`48` cases: `32` findings + `16` clean controls) • Selection: Best-of-`3` AST-only (`NFR01` / `NFR02` / `NFR06`)

LORA MULTI-AGENT (D) OVERALL	FINDINGS REFACTOR SUCCESS (D)	CLEAN CONTROL PRESERVATION (D)	7 SIGNATURE-SAFE TYPES (D)
91.67%	**87.50%**	**100.00%**	**28/28 (100.0%)**
vs Base Multi-Agent (B): +16.67 pp	vs Base Multi-Agent (B): +18.75 pp	vs Base Single (A): +25.00 pp	Excludes 4 `long-parameter-list` cases

1. Headline Comparative Results (Groups A–D on Frozen Test Split)

All four groups ran on a Kaggle Tesla T4 GPU with real `Qwen/Qwen2.5-Coder-3B-Instruct` inference (`--baseline base` , `--candidates 3` AST-only selection) and isolated sandbox verification (`0` `provider_unavailable` rows).

Group	Exec	Syntax	Tests Pass	Smell Res.	Refactor Success	Findings Success	Clean Control	Approved	pass@1	Latency	Tokens
A: Base Single	48/48	100.00%	77.27%	71.88%	**60.42%**	**53.12%**	75.00%	66.67%	77.27%	16.3s	541
B: Base Multi-Agent	48/48	100.00%	84.78%	84.38%	**75.00%**	**68.75%**	87.50%	77.08%	79.07%	22.4s	543
C: LoRA Single	48/48	100.00%	100.00%	18.75%	**45.83%**	**18.75%**	100.00%	45.83%	100.00%	66.7s	623
D: LoRA Multi-Agent	48/48	100.00%	100.00%	87.50%	**91.67%**	**87.50%**	100.00%	91.67%	97.73%	80.5s	669



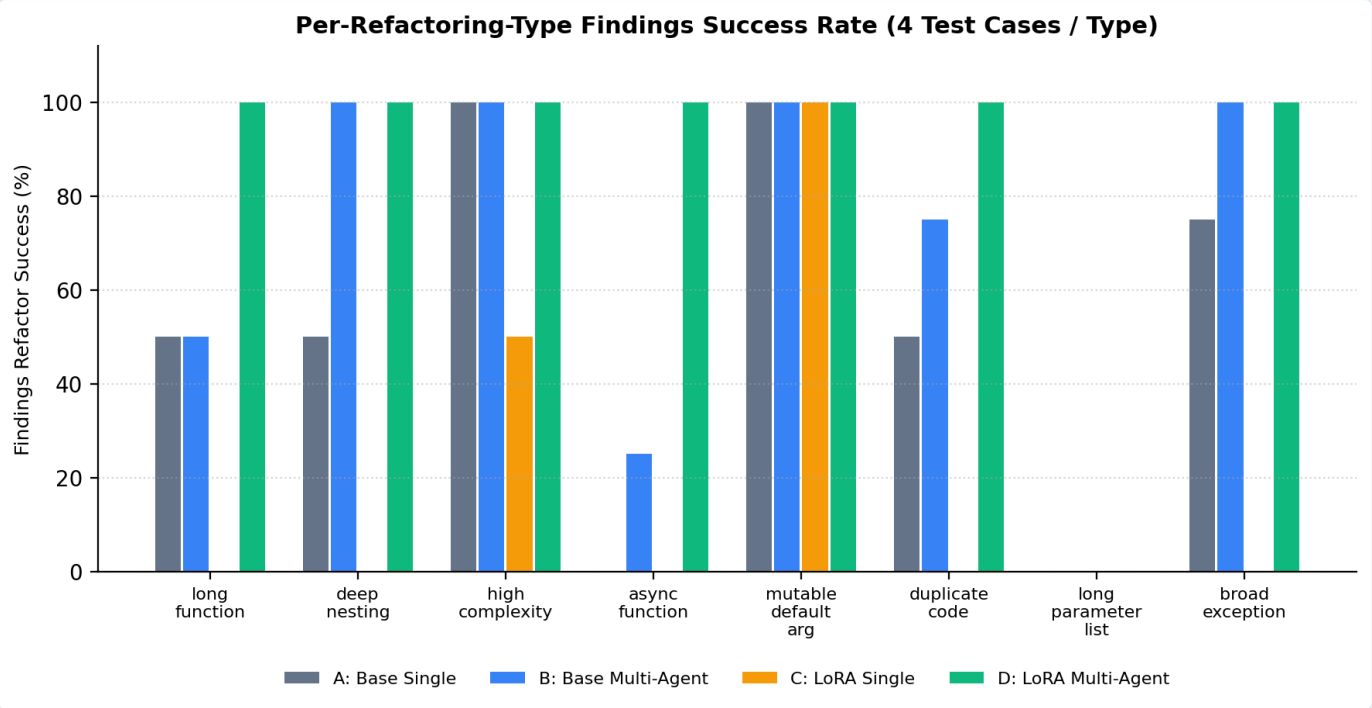
> **CPU Mock Reference vs. Real Untuned 3B Base:** Previously committed baseline numbers in `experiments/dataset_test_summary.json` (`refactor_success_rate = 45.83%`, `findings = 18.75%`) used deterministic CPU rules (`mock-refactor-v2`), not an LLM. This report compares the fine-tuned adapter (C/D) directly against the real untuned `Qwen2.5-Coder-3B-Instruct` model (A/B) under identical prompting and candidate budgets.

2. Fine-Tuning Provenance & Configuration

- **Base Model & Revision:** `Qwen/Qwen2.5-Coder-3B-Instruct` @ `488639f1ff808d1d3d0ba301aef8c11461451ec5`.
- **Actual Training Method: Native fp16 LoRA** (`load_in_4bit=false`, `optim=adamw_torch`, rank `16`, alpha `32`, dropout `0.05`, `all-linear`, `3` epochs, `max_length=2048`). *4-bit QLoRA remains unproven in this container and is explicitly distinguished from fp16 LoRA.*
- **Trainable Parameters:** `29,933,568` / `3,115,872,256` (`0.9607%`).
- **Kaggle Run Metrics:** Tesla T4 (`7157.5s`, peak allocated `9.31 GiB`), final validation loss (48 records, epoch 3.0) `0.005797`.
- **Verified Adapter Checksums:** `adapter_model.safetensors` SHA-256 `42ae18e5eb26976143077051824e97719222103d4352ca70a24ef1089bb8405f`; AST-only smoke check on `long-function-validation-01`: `syntax_valid=True`, `executed=False`.

3. Per-Smell Breakdown & Failure Diagnosis

Refactoring Type (4 findings cases each)	A: Base Single	B: Base Multi-Agent	C: LoRA Single	D: LoRA Multi-Agent
`long-function`	2/4 (50%)	2/4 (50%)	0/4 (0%)	4/4 (100%)
`deep-nesting`	2/4 (50%)	4/4 (100%)	0/4 (0%)	4/4 (100%)
`high-complexity`	4/4 (100%)	4/4 (100%)	2/4 (50%)	4/4 (100%)
`async-function`	0/4 (0%)	1/4 (25%)	0/4 (0%)	4/4 (100%)
`mutable-default-arg`	4/4 (100%)	4/4 (100%)	4/4 (100%)	4/4 (100%)
`duplicate-code`	2/4 (50%)	3/4 (75%)	0/4 (0%)	4/4 (100%)
`long-parameter-list`	0/4 (0%)	0/4 (0%)	0/4 (0%)	0/4 (0%)
`broad-exception`	3/4 (75%)	4/4 (100%)	0/4 (0%)	4/4 (100%)



Failure Mode Breakdown (All 48 Test Cases / Group)

Outcome / Root Cause	A: Base Single	B: Base Multi-Agent	C: LoRA Single	D: LoRA Multi-Agent
Passed (`refactor_success=True`)	29	36	22	44
Signature contract (`long-parameter-list`)	4	4	4	4
Sandbox test failed	8	7	0	0
Smell unresolved / AST no-op	5	1	22	0
Syntax invalid	0	0	0	0
Missing top-level symbols	2	0	0	0
Generation timeout	0	0	0	0
Other review rejection	0	0	0	0

Key Architectural Findings

- **Why Multi-Agent Coordination Matters Even After Fine-Tuning (Group D vs. C):** Under single-role evaluation (Group C), the harness passes `findings=[]` (no `AnalysisAgent`). Because the adapter was trained to return clean-

control code unchanged whenever `Detected issues:` is absent, Group C treats smell-bearing prompts like clean controls. Supplying AST findings plus bounded review (Group D) unlocks the learned refactoring behavior.

- `long-parameter-list` **Signature-Guard Contradiction (4 / 32 Findings = 12.5%):** The training dataset replaces 6 scalar parameters with a `@dataclass` request object, altering the call signature. Both `candidate_selection.py` (`-10,000` penalty) and `ReviewAgent._signature_violations()` unconditionally reject signature changes prior to behavioral execution. On the 7 signature-preserving refactoring types (`28` cases), Group D achieved **28/28** (**100.00%**).
- **Latency & Unmerged PEFT Inference:** `LocalModelProvider` evaluates best-of-3 generation sequentially through an unmerged `PeftModel`; merging weights or batching candidates is an inference-only speedup left for future work.

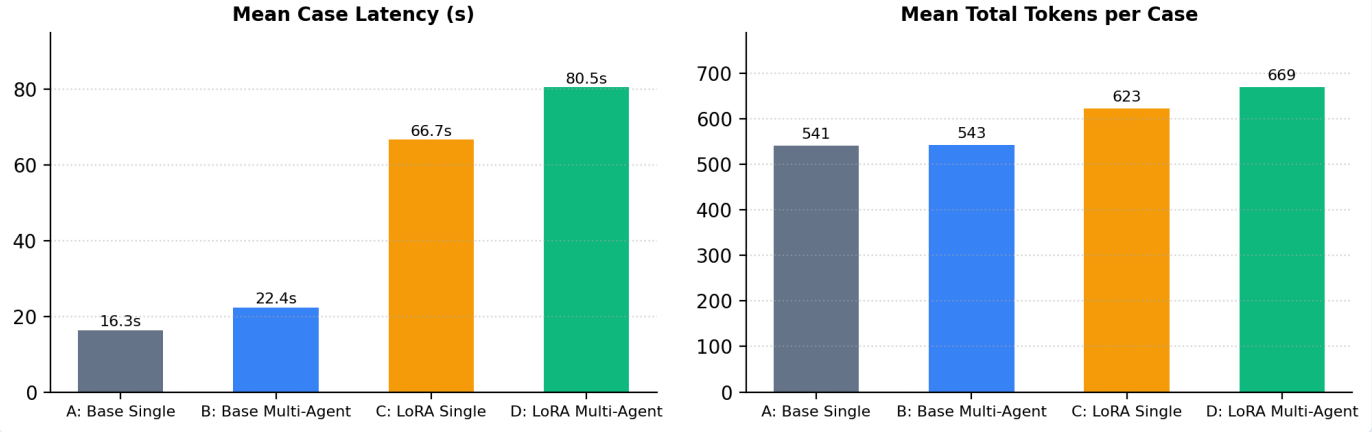
Representative Verified Fine-Tuned Refactoring (`deep-nesting-test-01`, Group D)

Original Code

```
def evaluate_deep_nesting_25(user, score, level, active):
    if user:
        if score > 35:
            if level:
                if active:
                    return user + score + level
    return 0
```

Refactored Output (Approved & Passed Sandbox)

```
def evaluate_deep_nesting_25(user, score, level, active):
    if user and score > 35 and level and active:
        return user + score + level
    return 0
```



4. Blind Human Evaluation & Release Verification

> **Blind Human Evaluation Status (NFR06):** The 4-candidate review packet (`experiments/review/packet.md` & `score_sheet.csv`) has been regenerated from `dataset_test_finetuned_results.csv`, but **human raters have not yet submitted scores**. Per NFR06, `scripts/aggregate_blind_review.py` refuses to fabricate or impute human scores.

- **Release Manifest:** `RELEASE_MANIFEST.json` covers `113` artifacts across `dataset`, `experiments`, and `frontend_dist` (aggregate SHA-256 `af2e0ed228db51a06d35986f6ed96ca5c6e1d61b518a89476eeecb71b30fe097`). All dataset and experiment CSV/JSON files were LF-normalized to match `.gitattributes` across Windows and Linux.
- **Automated Quality Gates:** check (114 files), test (110 pytest), docs (OK), dataset (416), contract (7 models), web (38 vitest + build), smoke (8/8), release (113 artifacts verified)

5. Limitations & Trust Boundaries (NFR01 / NFR02 / NFR06)

- **Parametric Dataset Scope:** `data/curated.jsonl` (416 records) is authored from 2–3 parametric templates per smell type; test-split gains measure in-distribution fidelity and do not claim arbitrary multi-file repository generalization.
- **Behavioral Verification Scope:** Auto-synthesized golden equivalence tests verify sampled inputs inside subprocess isolation (`PYTEST_DISABLE_PLUGIN_AUTOLOAD=1`, socket denial, POSIX rlimits) and do not constitute formal proof of equivalence.
- **Security Boundaries:** No submitted or generated code is ever executed in the FastAPI/harness process (AST-only inspection and candidate selection).